Maverick
MOTORSPORTS

DESIGN & ENGINEERING

PORTFOLIO

MU20 School of Opportunity | Lenovo | SeQR TMT BARS FOR SAFE HOMES | सेल SAIL | BeyondEdge deliver beyond boundaries | Prince Group | the earth prize | The Shishukunj International School Indore | SRT Shree Rapid Technologies | Corporate curate | Indore Talk | ABEC 3 5 7

DESIGN OBJECTIVES

*This page outlines Maverick Motorsports' **design objectives**, including performance goals, mass moment optimization, and **aerodynamic strategies**. It details **component-wise design choices**, and concludes with a **reflection table evaluating key design achievements.***

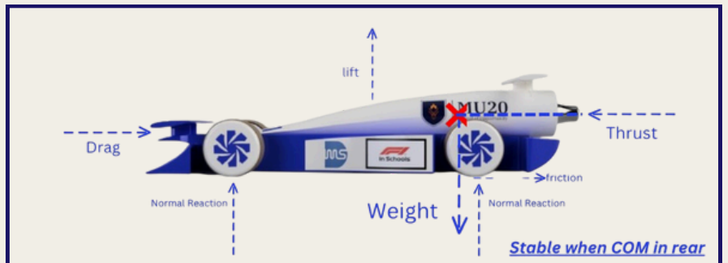
SUMMARY

DESIGN OBJECTIVES

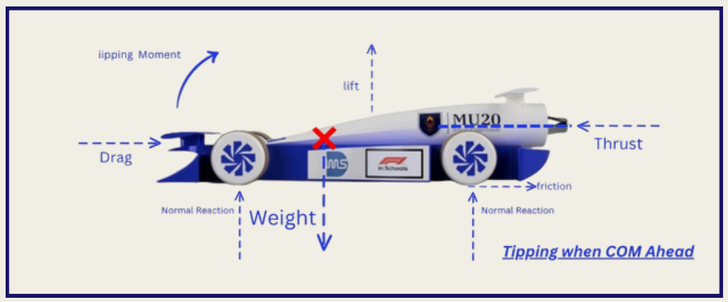
In our journey from regionals to nationals, we had some ideas in mind for the car's design and our team. Our ideology can be encapsulated into two parts:

1. Designed for Speed: Improve car race time(excluding the reaction training) from **1.3s to 1.1s**
2. Engineering for Perfection:
   a. Comply with all regulations to make a perfectly legal contender
   b. Create a durable design to minimise wastage

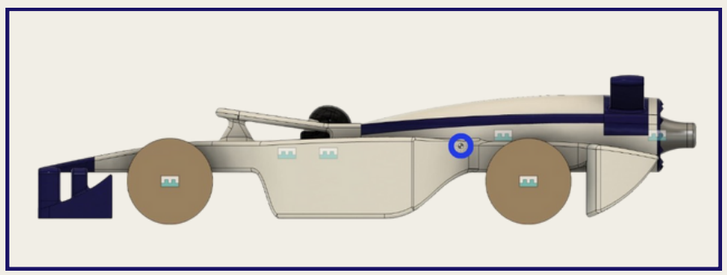
These basic objectives can be elaborated further under the following subheadings to achieve our aim:



Stable when COM in rear



Tipping when COM Ahead



COM of car

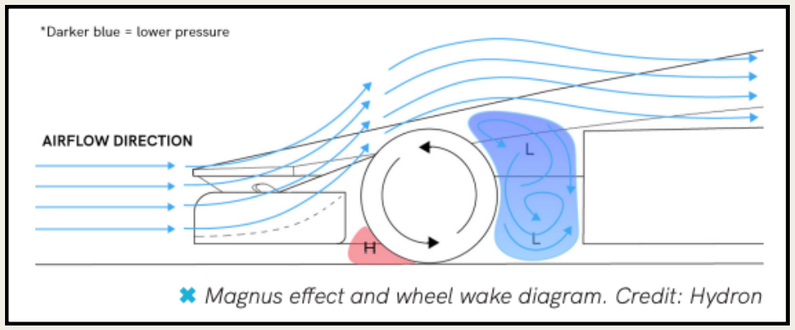
MASS MOMENT & THRUST OPTIMISATION

To prevent toppling and optimise thrust, the car's centre of mass (COM) should be placed slightly behind and in line with the thrust axis. This positioning balances torque during launch, prevents front lift, and ensures all thrust contributes to forward motion. A rearward COM also helps maintain momentum post-thrust by minimising deceleration and drag. Aiming for the minimum allowed weight(50g) is also essential to maximise acceleration.

CAR PART OBJECTIVES

Chassis, Sidepods and Diffusers:
The car body will be divided into two sections to optimise aerodynamics, reduce weight, and meet regulations. The top half will house the CO_2 canister and follow a tapered cylindrical profile with a gentle downward slope from rear to front, promoting smooth airflow and reducing pressure drag. The bottom half will be hollowed to reduce mass and create a low-pressure channel, enhancing diffuser performance and minimising underbody turbulence. Sidepods and diffusers will be seamlessly integrated with tapered edges to maintain attached, uninterrupted airflow and reduce vortex formation. This layout aims to ensure smooth transitions, a lightweight structure, and efficient flow control, forming the aerodynamic foundation of our car.

Front Wing
The front wing is the first point of contact with airflow, crucial for minimising drag and frontal area and optimising aerodynamics. A key function is reducing wheel wake, where rotating wheels create turbulence that disrupts downstream airflow. By directing air over the wheels, the front wing ensures cleaner airflow and improved efficiency. It also mitigates the Magnus effect, which causes uneven pressure distribution on spinning wheels, increasing drag and lift.



Magnus effect and wheel wake diagram. Credit: Hydron

By controlling airflow, the front wing reduces turbulence, enhances stability(by avoiding yaw instability), and maximises aerodynamic performance across the car.

Rear Wing
The rear wing's role concerning aerodynamic efficiency is that this is where the airflow has the highest tendency to detach from the rest of the car and create mini-eddies or vortices, which can slow the car down. Apart from this, the main aim was to minimise its frontal profile while being structurally sound and meeting all important regulations, as they have heavy penalties.

Wheels
The wheels play a key role in improving the car's aerodynamic efficiency by reducing air resistance and keeping airflow smooth and controlled around the lower body of the vehicle. The spokeless design not only increases structural strength but also allows air to pass cleanly through the wheel area, minimising turbulence and avoiding unwanted drag near critical aerodynamic components such as the rear wing and diffuser. This helps maintain balance and stability, especially at high speeds, ensuring that the car performs consistently under racing conditions. In addition, their lightweight construction and low moment of inertia contribute to quicker acceleration, improved energy efficiency, and better grip throughout the race, making the wheels an essential element of overall performance and control.

REFLECTION UPON COMPLETION OF DESIGN

Design Concept	Achievement Status	Description
Rearward COM	✅	Achieved through careful mass distribution & material selection
Effective Airflow at Front Wing	✅	Through Iterative Front Wing Design
Low Weight	✅	By creating a hollow body
Minimised Frontal Area	❌	Due to addition of Halo & Helmet
Reduced Turbulant Air at Rear	❌	Instead leveraged to provide propulsion using teeth

PRE-NATIONALS TESTING

*The page highlights our **testing of the Regionals' Contender** and testing of hypothetical concepts. This helped us build a base for our nationals development process. It also set the **basis of our design aims** and **evaluation process.***

SUMMARY

INTRODUCTION

Before commencing formal development of our Nationals contender, we conducted a focused round of pre-nationals testing to evaluate several hypothetical design concepts. Rather than beginning from scratch, we made controlled modifications to our regional car, Invictus, which served as our baseline data set. This allowed us to assess the performance impact of each concept—both aerodynamic and mechanical—under consistent testing conditions. The objective was to determine which innovations offered measurable improvements and were worth integrating into the final prototype for Nationals. This iterative process enabled data-driven decision making while minimising the risks of unvalidated design changes.

PHYSICAL TESTING & RACE PACE

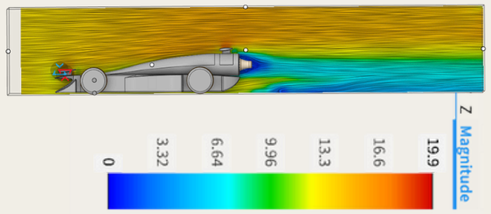
As a team based in Indore without access to a dedicated F1 in Schools test track, our best opportunity to gather real-world data was during the Regionals Finals. These races offered both race pace metrics and valuable observations under competitive conditions. Our car, Invictus (Variation 1), recorded a strong baseline time of **1.3 seconds** (excluding reaction time); however, our target for Nationals was **1.0–1.1 seconds.**

Post-race analysis revealed key aerodynamic inefficiencies. The front wing failed to direct airflow over the wheels effectively, with the aerofoil element above the nose instead deflecting air onto them, causing increased wheel wake and drag. This led to the development of Variations 2 and 3, which explored alternative wing geometries.

We also identified that the rear wing's fixed angle of attack in Variation 1 misaligned with the airflow, increasing drag. Variations 4 and 5 tested refined wing angles to better align with the flow and reduce resistance without compromising sensor triggering.

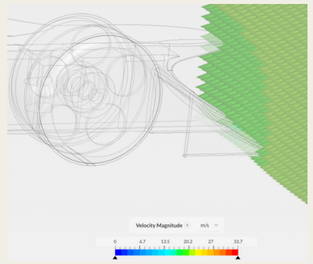
The following sections present the results and comparative analysis of these tests.





Race Time for Nationals

VARIATION 2



The first hypothesis we had was that the aerofoil wing shape we added on the front wing(for images, check page 3) was creating extra wheel wake, so we removed it and ran CFD Analysis.

It was evident that the wingspan was creating more problems than helping, and we needed to minimise the aerofoil's size to the minimum structurally stable, such that we could save on loss of speed to drag and also meet regulations. However, before we removed the idea completely, we wanted to see how a change in the shape would help. This test also reminded us to fix the aeroramps(results can be seen Tonitrum onwards on page 3)

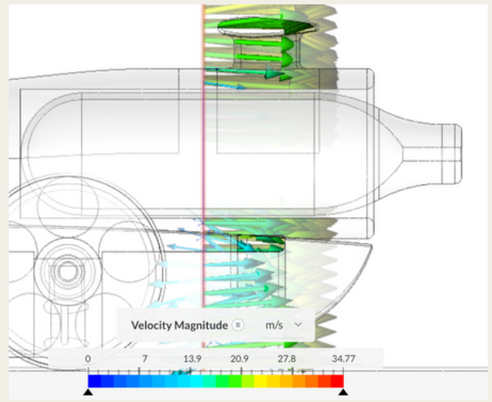
VARIATION 3

We tried to invert the wingspan to see if this would align the incoming airflow better.



Seeing negative effects on flow uniformity cemented our belief in a minimum viable front-wing size.

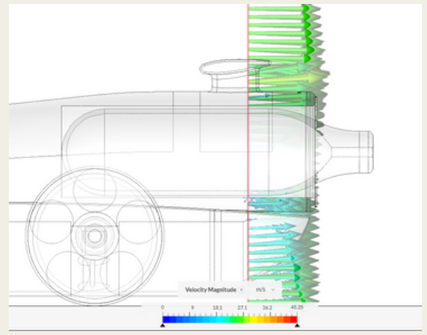
VARIATION 4



This variation was testing whether a lower angle of attack(straighter rear wing) would reduce the drag force on the rear wing. This test gave us positive results, but we also hypothesised a higher angle.

VARIATION 5

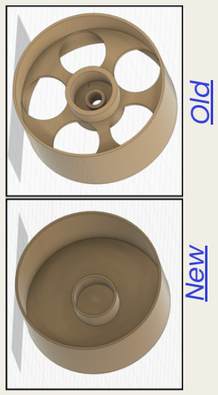
This variation had a higher angle of attack(more downturned rear wing).



Seeing a higher disruption in air flow uniformity and high drag force, we could determine that a lower angle of attack has better results; this ideology was translated into our national designs.

WHEEL CHANGES

Following the regional rounds, the wheels underwent further testing during which flaws such as high moment of inertia, excessive weight, and poor aerodynamic efficiency due to spokes were identified. As a result, a new set of wheels was designed, which aligned with the updated competition regulations and revised car design to ensure improved performance and compatibility.



Old

New

CAR DEVELOPMENT

*This page analyses the **Research and Development** done to create the **Chassis and Front Wing**. It also enlists how we **evaluated the design and how we made changes** on the basis of these evaluations. It also highlights **material choices.***

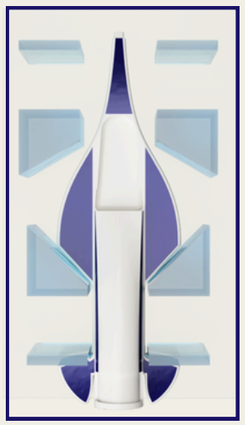
SUMMARY

ITERATIVE COMPONENT DESIGN

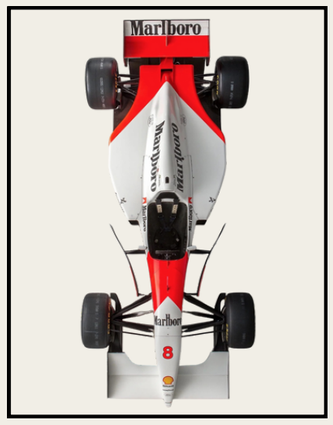
Chassis:



Our Chassis was heavily inspired by Ayrton Senna's MP4/8. Our rationale behind this was that in the old era, F1 vehicles largely relied on the body's aerodynamics to differentiate themselves from the field, whereas currently, most teams rely on strategies like flexible wings, engines, etc. This shape was also commonly seen in birds such as the Indian Swiftlet, with a wider midsection and a tapering downward nose. Apart from modelling the wheel regulations and meeting the minimum width criteria, our main chassis design had certain highlights in the design which were utilised to achieve performance.



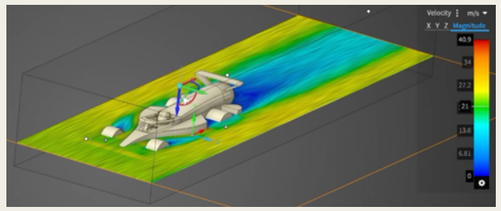
Car Body with Regulations Modelled



Inspiration

Tapered and Hollowed Sidepods:
The sidepod design, through multiple iterations, took a perfected shape such that the transition between the nose/front and the sidepods was a smooth one, allowing the turbulent air coming from the wheels to be deflected away from the car.

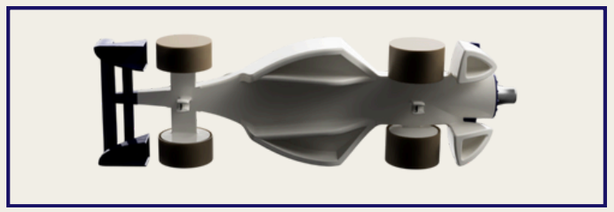
The side pods were also made hollow to minimise weight.



Side Pods and Diffusers Minimising drag due to turbulent air

Diffusers:
We carried forward the diffuser design from the regional finals with two small edits: we made the front wall slightly tilted instead of flat to allow the turbulent air from the rear wheels to be deflected away, and also made it hollow to allow for weight reduction, whose results are mentioned below.

Main Car Body and Halo:
The major goal in mind while designing the central part of the main car body was to ensure regulations were met and nothing obstructed the halo in the front, leaving spaces to join external parts securely, keeping the part behind the halo as close to a cylinder as possible to minimise frontal area and create a smooth face for the airflow and keep the floor as hollow as possible to minimise weight while maintaining minimum thickness and leaving space for the virtual cargo. This combined hollowing **took off 10g** from the chassis' weight.



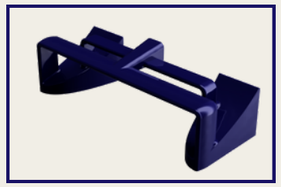
Floor of Chassis

Materials Used

Material	Density (g/cm3)	Tensile Str. (MPa)	Cost	Durability
Carbon Fibre	1.15	2500	Higher	High
Nylon 12	1.1	57	Mid Range	Moderate

We used Nylon 12 for the halo due to its lightweight and flexibility—it absorbs the impact from the stop mechanism without shattering. The helmet was made from carbon fibre to cut costs, as all carbon fibre parts were sponsored.

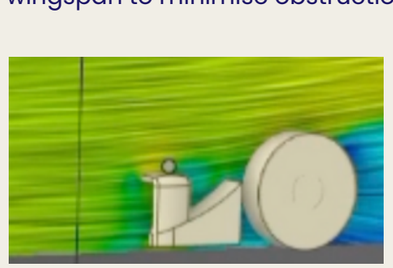
Front Wing:



Front Wing, Nose Cone and Support Structures

The Front Wing is the first point of contact between our car and the airflow. Skate ramps and snow shovels, which deflect objects upwards, were important inspirations for this design, as we wanted to achieve the same with the airflow.

The support structures are created to ensure structural integrity and avoid any breaks that may occur. The ramps on the side are responsible for shovelling airflow over the wheels and channelling the air in a way to maximise laminar flow, reducing the drag force and providing stability. We went for a minimal design to minimise drag, frontal area and weight. The support structures are positioned strategically to hide them behind the wingspan to minimise obstruction to airflow.





Materials Used

Material	Density (g/cm3)	Tensile Strength (MPa)	Cost	Durability
Carbon Fibre	1.15	2500	Higher	High
Nylon 12	1.1	57	Mid Range	Moderate
ABS plastic	1.04	40	Lower	Moderate

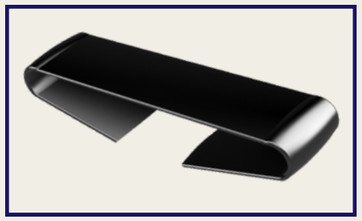
Carbon fibre was chosen for the wings due to its high tensile strength (2500 MPa) and low density (1.15 g/cm³). Compared to metals, carbon fibre offers excellent strength-to-weight performance, making it worth the cost.

CAR DEVELOPMENT

*This page analyses the **Research and Development** done to create the **Rear Wing, Wheel Support Structure, Wheels and Tether Guides**. It highlights **material choices.** A highlight here are the **innovative components** in our design highlighting our understanding and innovative thinking.*

SUMMARY

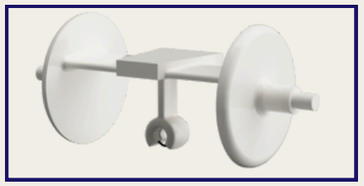
Rear Wing:



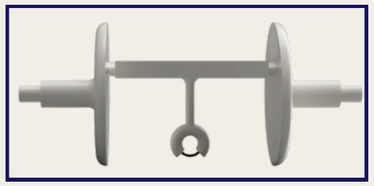
The Rear Wing is the final point of contact for external airflow to exit the car body. In Tempest, it is especially important for facilitating attached airflow as air vortexes exit the body and enter into low-pressure zones, creating mini-eddies that are trapped in our car's teeth.It is constructed in an aerofoil shape to reduce most of the drag that comes from the Halo and the Helmet, and has a negligible front profile. The angle of attack is aligned with the car body.

The material used is the same as our front wing, to ensure stability and balance lightness with flexibility.

Axle System:



Front Axle System



Rear Axle System

Due to differing wheel size regulations, the front and rear axle systems were designed separately. With hollow wheels doubling as hub caps, bearings were mounted on cylindrical ends, while an expanded section before each bearing prevented inward wheel slip. The axles provided sufficient surface area to ensure strong adhesion.

Materials Used

Material	Density (g/cm3)	Tensile Strength (MPa)	Cost	Durability
Carbon Fibre	1.15	2500	Higher	High
Aluminium	2.7	350	Mid Range	High
Nylon 12	1.1	57	Mid Range	Moderate

Nylon 12 was chosen for the axle system due to its lightweight, flexibility, and ease of manufacturing. Its flexibility also offered mild suspension and absorbed pressure during motion.

Wheel

Wheel Development



The regional wheels had internal spokes, which required hub caps to ensure smooth airflow and protect internal geometry, but this increased the overall weight and Moment of Inertia(MOI=197 at 14mm thickness). In response, the wheels were redesigned without spokes to reduce mass. Three prototypes with varied dimensions were tested, and the design was adjusted to meet the updated regulation mandating a minimum rear wheel thickness of **17 mm** and a front of **13 mm.**

FEA in Fusion 360 was used to validate structural integrity under mechanical stress and different loads.

The final front wheel recorded a **moment of inertia of 179** with a weight of **1.098 grams**, while the rear wheel showed an **MOI of 227 and weighed 1.334 grams.**

Materials Used

Material	Density (g/cm3)	Tensile Str.(MPa)	Cost	Durability	Young's Modulus
Carbon Fibre Reinforced Nylon	1.3	150	Higher	Very High	10
Ketron PEEK	1.31	115	Mid Range	High	3.6

PEEK was chosen for the updated wheels due to its high strength-to-weight ratio and CNC machinability, enabling precise, tight-tolerance profiles for high-performance use.

Tether Guide

For the tether guides, Fuji Silicon carbide rings were chosen as they offered an extremely low coefficient of friction and had a hardness of **9 on the Mohs scale.**

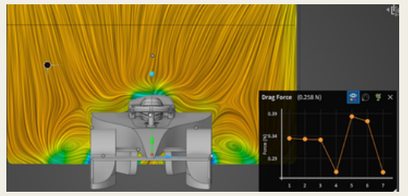
OUR INNOVATION/R&D

Tilted Sidepod

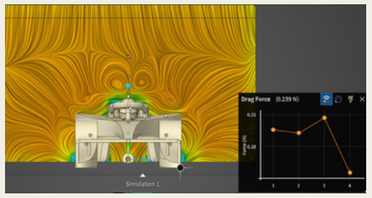




Our first innovation took inspiration from actual F1 vehicles(specifically SF 24), which feature a small dip or tilt in their sidepods to stabilise airflow. These tilted sidepods guide air outward & rearward, reducing pressure buildup, attaching & maintaing uniform flow and helping improve the efficiency of the central tunnel and diffuser, while making a smoother transition to the rearwings. The test results from our CFD analysis showed major improvements in the uniformity of airflow coming from the front wheels after the application of tilted sidepods.

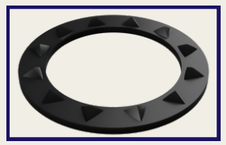


Before



After

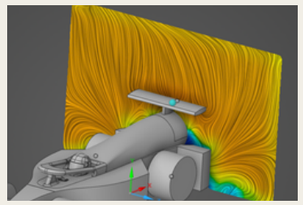
Teeth



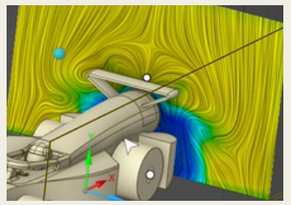


Small pyramid-like extrusions near the tail end of the car, surrounding the pressurised CO2 canister, were nicknamed teeth. Our inspiration in including this design came from one of the best drag racers to exist: The Honda S Dreamliner 2016.

These teeth help control the negative pressure drag that gets created in the rear of the car. Normally, the turbulent air left by the car is explosive and uncontrolled, creating a low-pressure zone, which has a negative effect; however, the teeth help convert this turbulence into organised eddies and vortices(which are used positively in this case), which turn back and crash into the car, further propelling it forward, converting something negative into something positive.



Before



After

CAR DEVELOPMENT

*This page offers a **clean and statistic based comparison** between our nationals rounds' prototypes and why and how was Tempest, the nationals contender, the best.*

SUMMARY

INVICTUS

Invictus was our regionals contender, created using older regulations lacking essential components such as the Halo, Helmet or Legal Ballast. The car was a heavier one, thus leaving significant space for improvement. Its tapered side pods, aeroramps(ramps on the front wing), diffusers, hollow central channel and material choices were some highlights. Efficient assembly mechanisms allowed us to go through the entire competition without needing repairs.

Evaluation:

The aeroramps were definitely on the right track, however needed some refinement in the slope to work better; an improved version was carried over to the final design. The tapered sidepod design was also improved and incorporated into the final design to allow for an even smoother transition. The diffusers were one of the highlights of Invictus and were superior to any other prototypes.



Statistics:

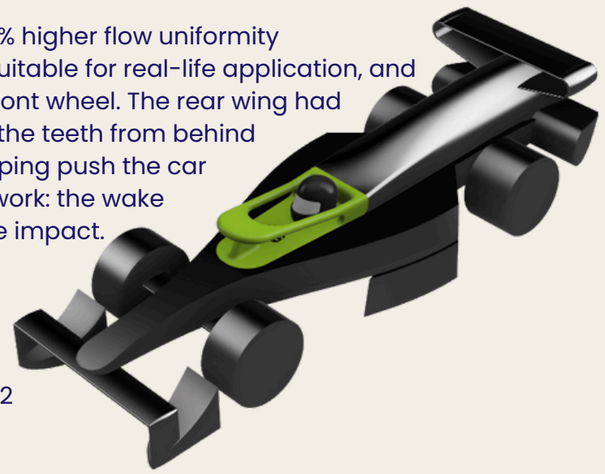
Frontal Area: 2053 mm2
Drag Force: 0.21N
Drag Coefficient: 0.32
Weight: 55g
Flow Uniformity-Factor: 0.913

NIMBUS

Nimbus is Latin for a thundercloud. Built to minimise area and increase straight-line speed, Nimbus was where we tested out two of our innovations: tilted sidepods and the vortex teeth. The front wing was kept consistent, as we believed its design was fairly optimised. We also made a change to the rear wing: by sweeping it around the body, we were able to cut down the area it was taking up. The diffusers were also modelled in an aerofoil shape to reduce the wake. It was significantly slimmer than the earlier prototype Tonitrum, but was ultimately deemed non-manufacturable.

Evaluation:

The tilted sidepods performed very well, giving us a 2% higher flow uniformity throughout the body. However, their shelling was unsuitable for real-life application, and also left far more space than necessary behind the front wheel. The rear wing had a much reduced drag factor, while visual analysis of the teeth from behind showed that eddies were being created and thus helping push the car forward. The diffusers were suboptimal and needed work: the wake beneath the diffusers was creating the most negative impact.



Statistics:

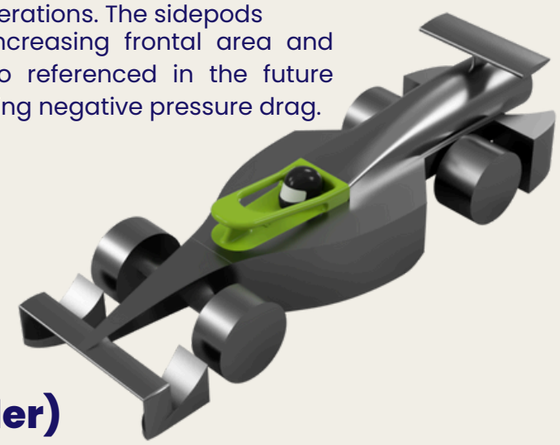
Frontal Area: 1759 mm2
Drag Force: 0.31N
Drag Coefficient: 0.4
Weight: 52g
Flow Uniformity Factor: 0.962

TONITRUM

Tonitrum is Latin for Thunder, as in the sound that comes in preparation for the Storm. This was the first prototype to finally incorporate the Halo and Helmet in line with the national regulations, leading to a sharp increase in the drag force. Incorporating a more hollow bottom, smoother side pods, a new diffuser design, and improved front wing design were important changes. These allowed for a predicted weight(after painting and finishing) of 50.5 grams.

Evaluation:

The improved front wing gave positive results, reducing front wheel wake by up to 80% and was used as a reference for further iterations. The sidepods began too early from the nose of the car, increasing frontal area and needed fixing. The new floor design was also referenced in the future processes. The diffuser design ended up increasing negative pressure drag.



Statistics:

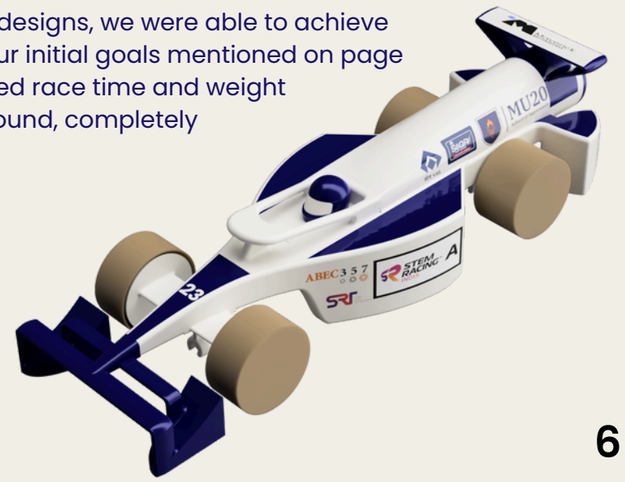
Frontal Area: 2063 mm2
Drag Force: 0.34N
Drag Coefficient: 0.5
Weight: 50.5g
Flow Uniformity Factor: 0.942

TEMPEST (Nationals Contender)

Tempest, Latin for a windy storm, represents our ideology of storming the track and unrestrained speed. Given the incorporation of the Halo and Helmet, the design had an extra source of obstruction; thus, in theory, its statistics should have been worse than Invictus, as in the case of the previous two prototypes. However, through extensive improvements in the front wing, chassis and material selections and innovations in tilted sidepods and teeth, we were able to bring about positive results.

Evaluation:

By combining the effective parts of the previous designs, we were able to achieve a stable and fast car. We were able to achieve our initial goals mentioned on page one, including minimal drag force, faster predicted race time and weight close to threshold. This design was structurally sound, completely legal and importantly aerodynamically efficient.



Statistics:

Frontal Area: 1899 mm2
Drag Force: 0.2N
Drag Coefficient: 0.34
Weight: 50.5g
Flow Uniformity Factor: 0.968

COMPUTER AIDED ANALYSIS

*This page offers insight into our usage of the Computer Aided Analysis Softwares. It talks about the usage of **FEA** to ensure structural integrity. It also mentions **how we set up CFD, tracking improvements and the initial results of CAA Testing.***

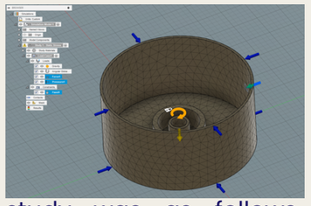
SUMMARY

FINITE ELEMENT ANALYSIS

With the wheel, reducing our MOI was concurrent with maintaining stability. Our aim was to understand whether the wheel could handle the stresses of being in a race, as well as external indirect factors (travelling, assembly, mishaps).

- We have added a radial load of 8N, considering external factors, and ensuring our wheel can bear the brunt of travel, carrying, and excess force. That's nearly 64 times the strength required.

- We also added a fluid penetrating pressure. This was done to simulate the pressure that would be generated by air hitting the surface of the wheels as the car raced ahead.



Since the density of air is 1.225kg/m3 at sea level, and the velocity of air in an F1 in Schools track is nearly 20 m/s, we can get a good estimate of the fluid pressure, making it about 245 Pa.

The standard procedure for the case study was as follows. After custom-setting the physical properties for Ketron PEEK (the material used for manufacturing our tyres), we followed these steps:

- Simplify geometry
- Set structural constraint on joining surface to axle
- Toggle gravity
- Set angular velocity
- Set tangential load of 8N throughout the wheel body
- Set a normal pressure load of 245 Pa throughout the wheel body
- Evaluate results

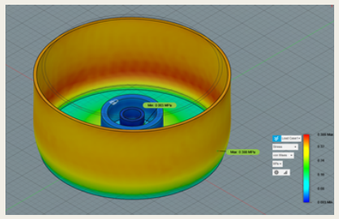
KETRON PEEK PROPERTIES

PROPERTY	VALUE	UNIT
Density	1,300.00	Kg m^3
Isotrophic Elasticity		
Derive from	Young's Modulus and Poisson	
Young's Modulus	3.8	GPa
Poisson's Ratio	0.375	
Bulk Modulus	5.0667E+09	Pa
Shear Modulus	1.3818E+09	Pa
Tensile Yield Strength	172	MPa

Custom-made Ketron PEEK as a material using engineering sources. All data was sourced from the technical datasheet of Mitsubishi Chemical Group

We ran three total tests on the wheels: Equivalent (von Mises) Stress, Shear Elastic Strain, and Total Deformation.

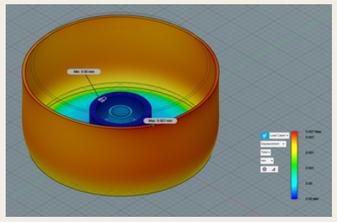
1. **Von Mises Stress**



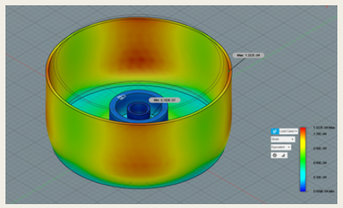
The von Mises Stress gave us a maximum value of 0.385 MPa. As a material, Ketron PEEK generally has a yield strength of 100-110 MPa. Therefore, compared to the tensile strength, the stress is almost negligible.

2. Equivalent Displacement

Equivalent displacement is a sum total of the displacements occurring in all three axes. The total displacement in the wheel's outer rim was not more than 0.002mm, which is almost negligible.



3.Equivalent Strain



Equivalent strain indicates a vector resolution of all the shear and normal strains on the body. The maximum strain was 1.33e-4, which is almost negligible.

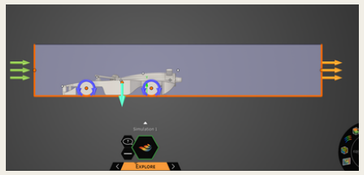
COMPUTATIONAL FLUID DYNAMICS

We used SimScale and Ansys Discovery to evaluate:

- Drag force and coefficient across variation
- Flow separation and wake behind wheels and the rear wing
- Pressure recovery at the diffuser
- Effectiveness of front wing flow redirection

Our Process, results and evaluation process is listed below:

BOUNDARY CONDITIONS



To set up the simulation, a certain number of boundary conditions are required to create the solution. The first part of these involves the fundamental Physics required to run the CFD:

- Gravity: 9.8m/s^2 in the negative Y-direction
- Turbulent Kinetic Energy: 1.95 m^2/s^2
- Kinematic Viscosity: 1.5e-5 m^2/s
- Air Density: 1.225kg/m^3
- Inlet Velocity: 22.5m/s
- Outlet Pressure (Gauge): 0 Pa
- Rotating Walls (on Wheels): 14,500 RPM (Revolutions per minute)

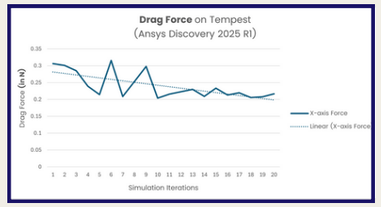
We set the walls as no-slip conditions, the X-Z plane as the grounding plane, the side and top walls as slip walls and created a flow region that was about 3-4 car lengths long in the rear.

TRACKING IMPROVEMENTS

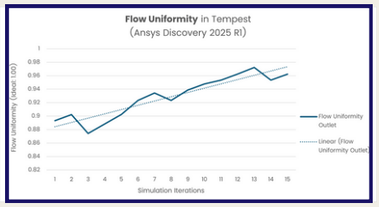
To track the effectiveness of our design changes, we made use of four main factors:

- Drag Force
- Air Velocity
- Flow Uniformity
- Pressure Drop

The drag force is an obvious indicator of the resistance the car experiences within the flow volume. Via measurements by CFD, our drag force was approximated to about 0.239N.



Drag force on Tempest



Flow uniformity in Tempest

Next, we derived the Drag Coefficient (C_D).

$$C_D = 2F_D / \rho \cdot v^2 \cdot A,$$

{F_D: Drag force on car (here 0.2N), ρ: Fluid density (here 1.225kg/m^3), v: Fluid velocity (here 22.5m/s), A: Frontal area of car (here 1899 mm^2/1.899e-6 m^2) }

This gives us a CD of nearly 0.34. Considering the addition of the Halo and the Helmet, this was an encouraging development.

This gives us a **CD of nearly 0.34**. Considering the addition of the Halo and the Helmet, this was an encouraging development.

COMPUTER AIDED ANALYSIS

*This page enters more in **depth of our CFD results**, It mentions **part by part analysis**, **improvements on the basis of CFD** and also our **Race Time Prediction** model.*

SUMMARY

Velocity is initially set at 22.5m/s, based on averages taken from previous F1 in Schools races. The ability of streamlines to maintain that velocity indicates the lack of flow separation, and thus, a good aerodynamic design.

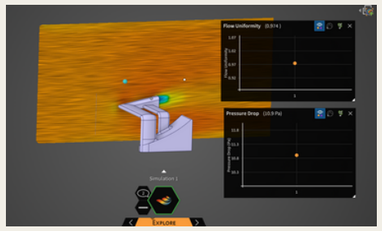
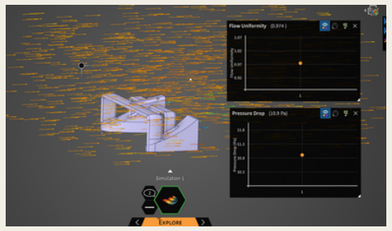
Via visual analysis, we were able to work on and improve on constraining our airflow velocity to about 24m/s ± 8m/s.

Flow Uniformity is a parameter ranging from 0.00 to 1.00, representing the percentage of retention of the airflow near the end. It indicates how much it has stayed attached to its original path of least resistance (i.e. the path it would have taken if the car body were an aerofoil); it is directly proportional to aerodynamic efficiency.

Pressure Drop indicates the difference in pressure from the inlet of the flow region to the outlet (where we've set Gauge Pressure as 0 Pa). A lower pressure drop indicates minimal energy loss (turbulent kinetic energy) and minimal flow separation. This helps us understand the deviation in overall airflow pressure; it is inversely proportional to aerodynamic efficiency. Our target value was about 10Pa for the rear wing and front wing, and 15Pa for the car body.

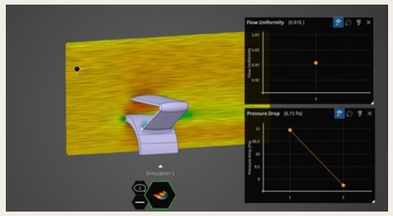
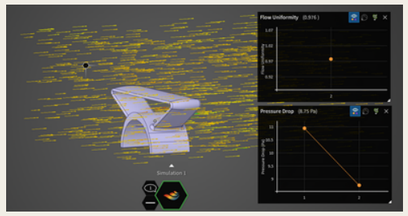
Part Improvements

Front Wing

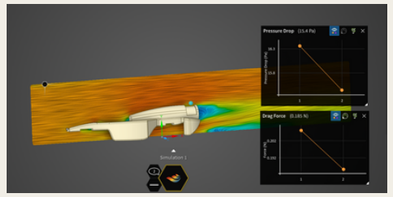
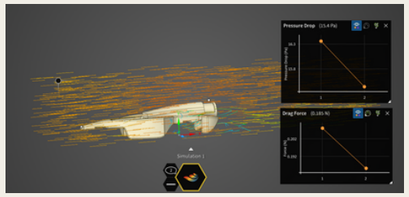
Problems with previous iterations- Higher mass; Premature flow separation; Regulations
Improvements- Built to regulations; Shelled as extensively as possible; Smoother ramps
Results and analysis- Better Flow Uniformity, Lesser Pressure Difference

Rear Wing

Problems with previous iterations- Too much frontal area; Wake creation
Improvements- Sweeping design to minimize area
Results and analysis- Better Flow Uniformity, Lesser Pressure Difference

Car Body

Problems with previous iterations- Too bulky; Wake near sidepods and behind front wheels
Improvements- Tilted sidepods; Smoother transition from nose to front of Halo; Directional loft to make rear end streamlined
Results and analysis- Better Flow Uniformity, Lesser Pressure Difference

Methodology Improvements

Mesh Optimisation :

Post studying various simulation models, we worked with the K-ω SST model. It is a hybrid turbulence model that combines the K-ω model near the wall (for accuracy in the boundary layer) with the K-ε model in the free stream (for robustness). The 'SST' stands for "Shear Stress Limiter." It prevents over-prediction of turbulence in adverse pressure gradients. This enhances stability and improves realism in the separation zone (i.e. near the teeth).
We aimed to improve the quality of our results; thus, we optimized the y^+ parameter carefully: it is a dimensionless parameter that represents the distance from the wall to the center of the first cell in the mesh near a wall.

$$u_\tau = (\tau_\omega / \rho)^{1/2},$$

$\{u_\tau$: Friction (shear) velocity created by wall, τ_ω: Wall shear stress, ρ: Material density$\}$

$$y^+ = u_\tau \cdot y / \nu$$

$\{y^+$: Non-dimensional wall distance, ν: Kinematic viscosity, y: Meshing cell height$\}$

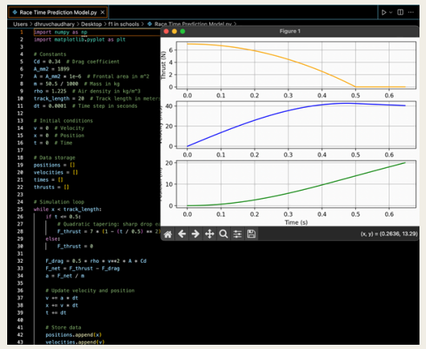
With the help of these algebraic identities, we were able to derive the fineness of our mesh. A y+ value of less than 1 indicates that the node lies inside the sublayer. It is particularly useful for heat transfer and resolving the boundary layer directly. Our y+ value came out to be approximately 0.5, indicating the mesh was almost perfect in its accuracy.
The major advantage that this model gave us was allowing us to adjust the fineness of our mesh to understand flow separation more accurately.

Racetime Prediction:

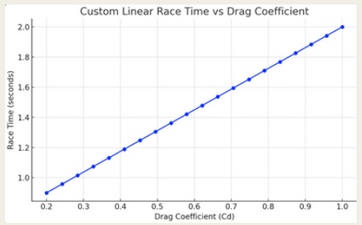
We used **Euler's integration model for 1D motion** with a variable force through a Python-based simulation. Applying Newton's Second Law of Motion:

$$F_{net} = m \cdot a \qquad F_{net} = F_{thrust} - F_{drag}$$

We can determine the value of velocity and distance covered in each time step, and this can be calculated to analyse the time taken to cover the track (20m). Initial thrust subsides after about 0.7 sec (calculated using the potential energy of the cartridge and the flow continuity equation). With a time step of 0.0001sec, a drag coefficient of 0.34N and a 1899 mm^2 frontal area with a peak thrust of 10N, we got a predicted race time of **1.06 seconds.**



Script Running

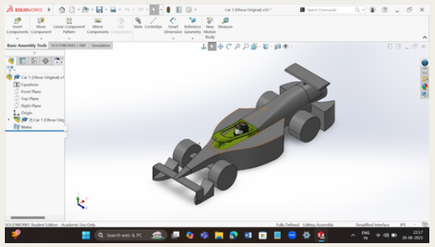


Result

3D MODELLING

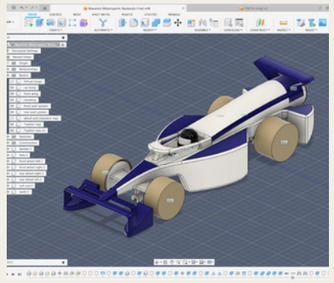
*This page highlights the tools and **methods we used for our 3D Modelling**. It specifies f**unctions used, Quality of surface analysis** and Taking into account **manufacturability.***

SUMMARY



Our design journey revolved around iterative 3D modelling, combining form, function, and manufacturability. We utilised both SolidWorks and Fusion 360 to bring our concepts to life.

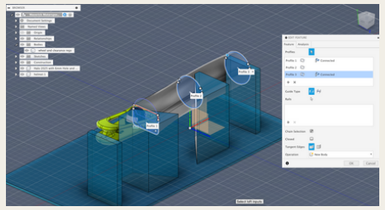
Our early prototype, Tonitrum, was developed in SolidWorks due to limited access—only one design engineer on our team had the license. Despite being a solo effort, this prototype allowed us to experiment with chassis proportions, basic aerodynamic shaping, and component packaging. It served as a critical stepping stone for concept validation and team discussions. For the final prototype, we used SolidWorks for its advanced rendering platform.



As the design matured, we transitioned fully to Fusion 360, primarily for its cloud-based sharing and real-time collaboration capabilities. This allowed multiple members to contribute simultaneously and enabled streamlined feedback integration from technical mentors.

CORE MODELLING TECHNIQUES USED

- **Revolve**: Used to generate symmetrical aerodynamic profiles for the Tether Guide Slots
- **Extrude**: Applied to bring sketches to life and providing appropriate shape to the car.
- **Shell**: Used to hollow the underside and reduce mass while preserving structural integrity.
- **Loft**: For the creation of the body, we used Directional Loft to adjust the angle at which the body was connected. To prevent the car body from increasing its lateral area (thus, less aerodynamically efficient), we used G2 Loft (Curvature Continuity).
- **Sweep:** Used in both the wings to create the main span. We used path sweeps as we did not have complex features needing guiderails.
- **Combine:** Used to join mirrored components or components created separately like rear wing span and joining ring.
- **Constraints:** To ensure precise sketching and easy editing later on. We extensively used dimension, parallel, horizontal and vertical constraints.



Loft In Action

- **Fillet:** For long edges (for example, edges alongside pods, main body etc.) we used Rolling Ball fillet with a radius range of 0.75mm to 3.5mm depending on the part filleted. In places where joints were made (for example, wheel system joining points near the front wing), we used a Setback fillet.
- **Decals and Canvas:** To add logos and elements of the livery.
- **Join:** For assembly simulations.. We made use of Rotating Joints for the wheels and Rigid Joints for the axles to better mimic the stress conditions during the race.

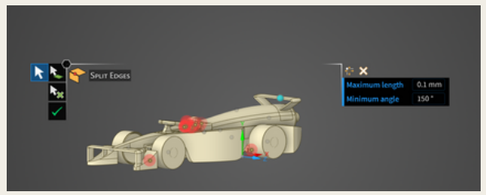
ADVANCED CAD FUNCTIONS

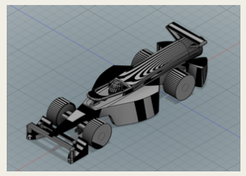
- **Parametric Constraints & Timeline Management:** Allowed us to control dependencies and edit features without affecting downstream geometry.
- **Assembly Workspace:** Helped validate tolerances and interference for moving parts like wheels and axles.
- **Component Isolation:** Enabled testing and redesigning parts like the rear wing or axle casing independently.
- **Material Simulation (Early FEA):** Helped estimate stress points in lightweight regions before exporting for detailed analysis.
- **Construct:** Used to create sketches in different planes and places.

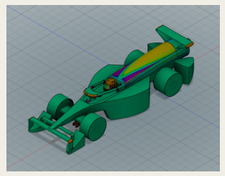
This modelling foundation helped us prototype with high fidelity and ensured we maintained regulation compliance (e.g., minimum thickness, canister clearance, tether slot alignment) at every stage.

GEOMETRY REPAIR AND SURFACE ANALYSIS

The first step of CFD involved preparing our geometry for analysis. By using the Spaceclaim 2025 R1 features within the Ansys suite, we repaired certain bits of our geometry:our precheck involved looking for stitched edges, existing gaps, interferences, missing faces, and duplicate curves. We then manually repaired each inconsistency. One issue we ran into was the simplification of the CAD geometry.



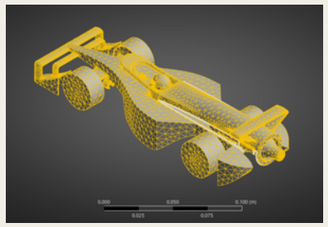




We solved this in two ways: combining all of the bodies into a single body for the entire car, and then using a STEP file instead of the usual.STL, we were previously using.

We conducted a Zebra Analysis and Curvature Map Analysis in Fusion 360 to understand the manufacturability and feasibility of our design. This was done to ensure there were no illegal intersections in the car's faces and bodies. Since all the curves are non-intersecting, that shows that the curvature of our car is within the legal tolerances and limits of CNC machining.

MESHING



We used a Hex-Dominant mesh throughout our car body (pre-filleting) to ensure that orthogonality is maintained, especially in cells near the walls of the body and the diffusers.

MANUFACTURING CONSIDERATIONS

During 3D modeling, we took several measures to ensure our designs were manufacturable using the available processes and tools. We added **functional tolerances between all interfacing parts,** following the principle that excess material can be removed during post-processing, but missing material cannot be added back. For CNC machined parts, we **avoided sharp internal corners and added fillets that matched the cutter's bur size**, which helped prevent tool strain and improved machining speed. We also ensured all CNC parts maintained a **minimum thickness of 3.5 millimeters** to avoid flexing or breakage during milling. For components like axle mounts that required precise attachment holes, we used a method involving **external shelling:** creating a 1 millimeter shell around the inner part and using it to cut out a cavity in the outer body. This approach gave us clean fits and consistent alignment. In the case of 3D printed parts, we designed every feature to stay **within a 45 degree overhang** limit to avoid support structures and ensure smooth printing. **Wall thicknesses** were chosen based on the print process: **0.8 millimeters for SLS** to keep the parts lightweight yet sturdy, and **1.5 millimeters for FDM** to ensure mechanical strength and reliable prints. By integrating these constraints early in the design phase, we saved time in manufacturing and avoided costly revisions.

Car Developement Process

MANUFACTURING

*This page talks **technologies used for manufacturing**, **machines chosen** and rationale. It goes into depth of the **CAM and CNC** process and our experience.*

SUMMARY

GOALS

1. Minimise manual finishing
2. Reduce the chance of defects by taking enough precautions and leaving adequate tolerances
3. Ensure that the car's assembly is firm

MACHINES USED FOR 3D PRINTING

EOS Formiga P110 & sPro 60



We selected the EOS Formiga P110 for its 200 × 250 × 330 mm build volume and 0.06 mm layer resolution, which provided the fine detail and surface finish needed for the Nylon 12 axle components. The sPro 60's larger build capacity and high-volume production ensured we could produce multiple strong, dimensionally stable parts efficiently. Both machines delivered the accuracy, strength, and consistency required for the axle system. Their proven performance with Nylon 12 material also minimized warpage and ensured reliable mechanical properties for load-bearing applications.

Raise3D E2CF

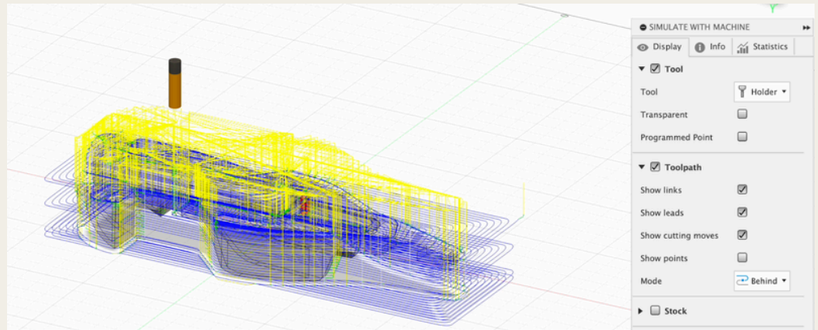


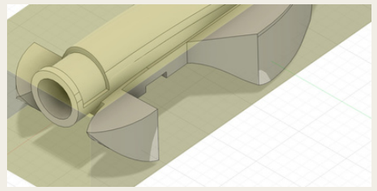
The Raise3D E2CF was used to manufacture carbon fibre-reinforced parts such as the front wing, rear wing, and helmet. It was chosen for its 330 × 240 × 240 mm build volume, 0.1 mm layer height, and ±0.1 mm dimensional accuracy, which allowed us to produce precise and lightweight components. Its hardened steel nozzles and reinforced feeding system ensured smooth handling of abrasive carbon fibre filament. Additionally, features like a fully enclosed chamber, heated bed up to 110°C, and dual independent extruders helped maintain consistent print quality and structural integrity throughout.

CAM & CNC

Usage of Fusion 360 Manufacturing Workspace:

For the national round, the Fusion 360 Manufacturing workspace was used extensively to replicate the CNC machining process as accurately as possible. The CNC tool used by our manufacturing partner, Stemplify, was recreated within the software environment, helping us simulate real-world conditions. We generated a complete 3-axis toolpath covering all critical surfaces and features of the car body. The stock dimensions were modelled directly using official STEM Racing model block specifications to maintain manufacturing compliance. Tool clearances and cutter trajectories were closely analysed and refined to ensure no collisions or incomplete cuts would occur during actual production. This simulation enabled us to iterate and refine designs with confidence, saving both time and material in later stages.





One of the final simulations revealed a critical flaw in the car body. The overall width exceeded the model block dimensions by 0.5 millimetres on both sides. This would have resulted in machining complications, as the design would not conform to the available stock dimensions specified under STEM Racing regulations. Early identification of this issue through CAM validation allowed the team to modify the design accordingly and bring the width within the permissible 65 millimetre limit. This proactive correction prevented potential delays and material loss during manufacturing.

Overall, this experience significantly deepened our understanding of CAM and its importance in product development. We learned how digital toolpath simulation can expose real world machining problems well before production. It taught us to view design through a manufacturing lens, focusing not just on geometry and performance but also on feasibility, stock fit, and cutter behaviour. The adoption of CAM helped us bridge the gap between design and execution, enabling a more reliable, efficient, and production ready workflow.

Model Dimensions		Model Dimensions	
Model Width (X)	194.566 mm	Model Width (X)	193.948 mm
Model Depth (Y)	65.9132 mm	Model Depth (Y)	41.69 mm
Model Height (Z)	42.5648 mm	Model Height (Z)	64.0803 mm
Before		**After**	

During the regional stage, CAM and CNC analysis software were not utilised in the development process. As a result, several design aspects failed to meet manufacturing regulations, leading to significant mark deductions. The absence of toolpath simulation, stock dimension validation, and material removal analysis resulted in parts that were not manufacturing-ready. Recognising this gap, we made a major shift in our workflow for the national round by integrating CAM software into our design pipeline. This allowed us to simulate tool paths, test machining strategies, and ensure that every design was dimensionally accurate and compatible with CNC machining standards from the start.

CNC MACHINES USED

Cosmos CVM 800



The Cosmos CVM 800 was selected for machining Ketron PEEK wheels due to its 800 × 500 × 500 mm working area and spindle speeds up to 8000 rpm, which supported precise cutting of complex wheel geometries. Its positioning accuracy of 0.01 mm and repeatability of ±0.003 mm ensured consistent concentricity and surface finish across all prototypes. The 24-tool automatic changer allowed efficient multi-step machining without manual interruption.

Denford MRC 40



The Denford MRC 40 was used to machine the main car body due to its 500W spindle motor, 110 mm Z-axis travel, and large working envelope, which made it ideal for detailed 3D contouring. Its high-speed performance and consistent dimensional accuracy ensured that even complex curves were reproduced with precision. The rigid frame reduced vibrations during machining, improving surface finish and tool control. Its quick tool change capability also allowed efficient transitions between operations, making it well-suited for cutting the entire car body accurately and reliably.

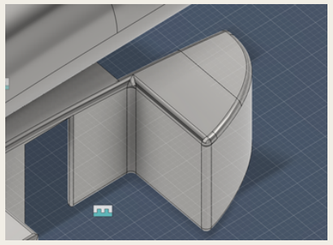
MANUFACTURING

*This page talks about the **Assembly and Final finishing process** including surface finishing, painting, glueing etc. It also mentions **Workplace Safety** measures and **Quality Analysis:** How we ensured our car model met our standards.*

SUMMARY

MANUFACTURING REFORMS

At the regional stage, no tolerances were applied for CNC machining or 3D printing, resulting in poor fits and extensive manual sanding during assembly. This increased post-processing time and introduced risks of dimensional inaccuracies. For the nationals, we referred to the specific tolerance limits of each machine model and incorporated clearances accordingly. This ensured better part compatibility, reduced manual work, and improved overall manufacturing accuracy. The regional car body design included extremely thin fillets and narrow joints, making it structurally weak and prone to damage during CNC machining. In some cases, this could have caused the model block to crack or break. For the updated design, we increased joint thickness and used smoother transitions to strengthen the structure and ensure safer, more reliable manufacturing.



ASSEMBLY & WORKPLACE SAFETY

Outsourced materials:

The axle system and hub caps, made of Nylon 12, were outsourced to Makenica since our main manufacturer, Stemplify, did not support this material. The outsourcing also helped reduce costs due to Makenica's competitive pricing.

The front and rear wings, made of carbon fibre, were provided free of cost by our aid sponsor, Shree Rapid Technologies, who had access to the correct grade suitable for our use.

The wheels, made from Ketron PEEK, were outsourced to Micro Tools and Services. As PEEK is a specialised, imported plastic, it requires highly precise VMC machining with very low tolerances, which the vendor in India successfully carried out.

Sanding & Finishing:

After CNC machining, the car body was sanded to ensure a smooth and even finish. Starting with 320-grit sandpaper to remove surface defects, we progressed to 2000-grit for final polishing. An electric hand sander was used for precision and consistency, especially in detailed areas.

Carbon fibre and Nylon 12 parts, made via 3D printing, also had surface irregularities and uneven edges, which were manually sanded for improved fit and finish. Although the VMC-machined wheels were highly precise, minor edge inconsistencies were corrected using 1000-grit sandpaper to ensure smooth rotation and balance.

Painting & Sealing:

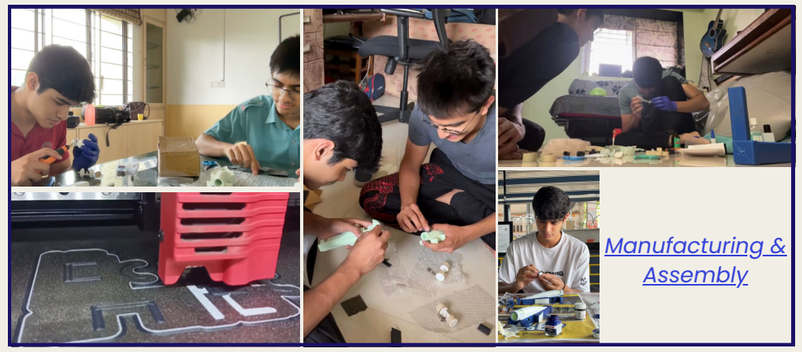
- Each part of the car was painted individually to reduce weight by avoiding hidden areas and to ensure greater precision. The entire primer and paint process was carried out by a professional company, Lords of Detailing, equipped with specialised tools for a smooth and uniform finish.
- Before priming, the car body was sealed with a mix of Fevicol and water to prevent polyurethane foam from reacting with hydrocarbon-based primers and paints. A coat of plastic primer was then applied to all parts and left to dry for 2 hours, chosen for its strong adhesion and protection against corrosion and cracking.
- Spray paint was applied according to the colour scheme, using masking tape to separate sections. After drying, waterslide decals formed the livery, and vinyl stickers were used for the 'STEM Racing' logo, as per regulations. A clear coat was added to seal the surface and protect it from moisture. No paint or primer was applied to attachment spaces, as they were dimension-critical.

Assembly:

After painting, Loctite SF 770 primer was applied to all joints and left for 1 minute to activate the inert Nylon 12 surfaces. Loctite 406 was used for larger joints, while Loctite Precision was applied to small gaps like wheels and bearings. This ensured strong, reliable bonding. Attachment spaces were included in the car body, with space left for glue to securely fix outsourced parts like the front wing, rear wing, and wheel system.

WORKPLACE SAFETY:

Risk	Risk Cause	Risk Control
Skin irritation, Breathing problems	Spilling of super glue, inhalation of spray paint	Medical masks and high-grade gloves were used
Hand injuries and cuts	Accidental contact with sanding surfaces or sharp parts	Use of clamps and blocks to hold components securely while sanding
Eye injury	Flying dust or particles during sanding or drilling	Use of protective safety goggles at all times during finishing processes



Manufacturing & Assembly



Physical Testing

QUALITY ANALYSIS

Without access to a regulation F1 in Schools track, our team had to approach physical testing through alternative yet effective methods. Our process was centred around three major components.

First, we rigorously monitored and **adjusted the car's weight** throughout development to ensure it consistently met the target, a crucial factor for maximising acceleration and maintaining structural efficiency. Considering we needed only minor fine-tuning, we opted **not to use the legal ballast and relied on controlled sanding and painting** to adjust weight.

Second, we conducted in-depth **physical scrutineering using precision tools**(vernier callipers, gram weight scale, screw gauge) to confirm that all critical dimensions, angles, and tolerances matched our CAD model, reducing the risk of disqualification and ensuring aerodynamic accuracy.

Finally, to validate our digital simulations, we collaborated with SGSITS, our local engineering college, to carry out a **wind tunnel test.** This allowed us to compare our CFD predictions with real-world airflow behaviour, giving us valuable insight into the performance of our aerodynamic features and enabling final refinements to the car design.